GitHub link: https://github.com/poluananya13/LinkedIn_Data_Set

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A Comparative Analysis of Support Vector Machine Kernels for Job Classification

Abstract:

This document analyzes the efficiency of Support Vector Machine (SVM) kernels in a job classification assignment utilizing a real-world dataset comprised of 811 records with skill-related features. The main goal is to assess the impact of various kernel functions on the performance of the model. Three SVM models were developed using Linear, Polynomial, and Radial Basis Function (RBF) kernels.

The dataset underwent preprocessing where we selected numerical features and applied a train-test split for model evaluation. We measured performance using accuracy metrics, classification reports, and confusion matrices to offer both overall and class-level insights. The results show that the Linear kernel achieved perfect classification accuracy, indicating that the dataset is mostly linearly separable. The RBF kernel also performed well, while the Polynomial kernel displayed much lower performance, which suggests overfitting and sensitivity to class imbalance. These findings emphasize that more complex models do not always result in better performance. The study offers important insight into the need to choose suitable machine learning models based on data characteristics, showing that simpler models can often work better for structured datasets.

1.Introduction:

Machine learning has become an essential tool for analyzing large datasets and making data-driven decisions in various fields. Among the many supervised learning methods, Support Vector Machines (SVM) are popular for classification tasks because they can create optimal decision boundaries between different classes. A key strength of SVM is its use of kernel functions, which help the algorithm manage both linear and non-linear relationships within data. By changing the input space into higher-dimensional representations, kernels enable SVM models to effectively separate complex datasets that cannot be separated linearly. This report focuses on evaluating the performance of different SVM kernels in job classification. Using a real-world dataset that includes job-related features and skill indicators, we implement and compare three kernel functions: Linear, Polynomial, and Radial Basis Function (RBF). The goal of this study is to examine how kernel selection affects classification performance and to find out if more complex kernels yield better results. We evaluate the models using standard performance metrics, such as accuracy, classification reports, and confusion matrices. Through this analysis, the report aims to provide insight into the effectiveness of SVM kernels and emphasize the importance of choosing the right machine learning models based on the dataset's structure and characteristics.

2.Dataset Description:

Machine learning has become a key tool for analyzing large datasets and making decisions based on data across various fields. Support Vector Machines (SVM) are popular among supervised learning methods for classification tasks because they can create optimal decision boundaries between different classes. A major advantage of SVM is its use of kernel functions, which help the algorithm handle both linear and non-linear relationships in data. By transforming the input space into higher-dimensional representations, kernels enable SVM models to effectively separate complex datasets that cannot be separated linearly. This report focuses on evaluating the performance of different SVM kernels in job classification. We use a real-world dataset containing job-related features and skill indicators to implement and compare three kernel functions: Linear, Polynomial, and Radial Basis Function (RBF). The goal of this study is to see how kernel selection impacts classification performance and to determine if more complex kernels provide better results. We assess the models using standard performance metrics like accuracy, classification reports, and confusion matrices. Through this analysis, the report aims to give insight into the effectiveness of SVM kernels and highlight the importance of selecting the right machine learning models based on the dataset's structure and characteristics.

3.Methodolgy:

The study follows a structured machine learning workflow to evaluate SVM kernel performance. First, the researchers preprocessed the dataset by selecting numerical features and excluding categorical variables to make it compatible with the model. Since there were no missing values, no imputation was needed.

- Numerical features were selected for training
- The dataset was split into 80% training and 20% testing sets
- Three SVM models were trained using:
    - Linear kernel
    - Polynomial kernel
    - RBF kernel

➢ Model performance was evaluated using:

- Accuracy
- Classification report
- Confusion matrix

4.Results:

4.1Accuracy Comparsion:

The comparison of the three SVM kernels shows clear differences in performance. The Linear kernel reached the highest accuracy of 100%, meaning it perfectly classified all instances. The RBF kernel also did well, achieving an accuracy of about 98%, with only a few minor misclassifications. On the other hand, the Polynomial kernel performed much worse, with an accuracy of around 60%. These results indicate that the dataset is mostly linearly separable, which helps the Linear SVM perform at its best. The poorer performance of the Polynomial kernel suggests that adding complexity to the model can hurt performance, especially when the data does not need non-linear changes.

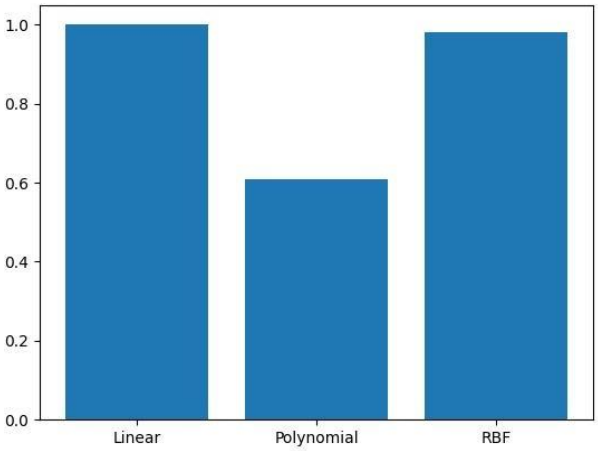

Figure 1: Accuracy Comparison of SVM Kernals

4.2 Class Distribution

The distribution of classes within the dataset highlights an uneven representation across various job categories. Class 4 holds the largest number of instances, whereas Class 2 has the smallest. The other classes are distributed somewhere in between these two extremes.

This skew is crucial as it can impact the performance of models, especially those that are more complex. In this analysis, the Polynomial kernel demonstrated a significant bias towards the majority class. Conversely, the Linear and RBF models managed to better address the imbalance.

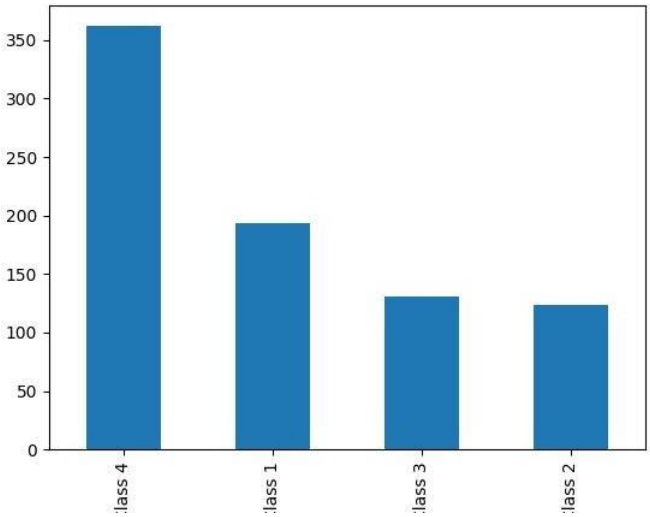

Fig2 :Class Distribution

4.3 Confusion Matrix – Linear Kernel

The confusion matrix for the Linear SVM model indicates flawless classification performance. All predicted values are perfectly positioned along the diagonal, demonstrating that every instance has been correctly classified into its appropriate class. There are no values outside the diagonal, signifying that there were no misclassifications across any of the classes. This validates the model's capability to entirely separate the data using a linear decision boundary. The findings imply that the dataset is extremely linearly separable, enabling the Linear kernel to perform at its best without needing more intricate transformations. This also clarifies why the Linear SVM achieved an accuracy of 100%.

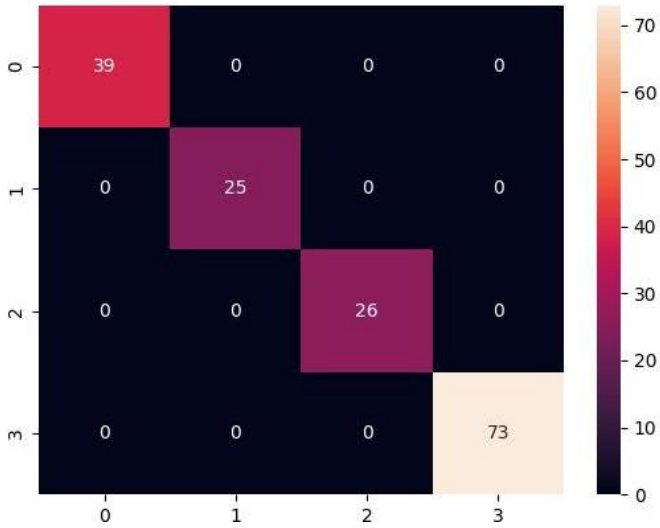

Figure 3: Confusion Matrix – Linear Kernel

The Linear model shows perfect classification, with all predictions correctly placed on the diagonal

4.4 Confusion Matrix – Polynomial Kernel

The confusion matrix for the Polynomial SVM model indicates subpar classification performance when compared to the Linear and RBF kernels. A significant number of instances are misclassified, with most predictions largely skewed towards Class 4, which is the predominant class in the dataset. This results in minimal correct predictions for Classes 1 and 2, suggesting that the model has difficulty distinguishing between various categories. The presence of substantial off-diagonal values emphasizes the high degree of misclassification. The findings imply that the Polynomial kernel introduces excessive complexity, causing overfitting and reducing generalization capability. Additionally, it shows a high sensitivity to the existing class imbalance in the dataset, adversely affecting its performance.

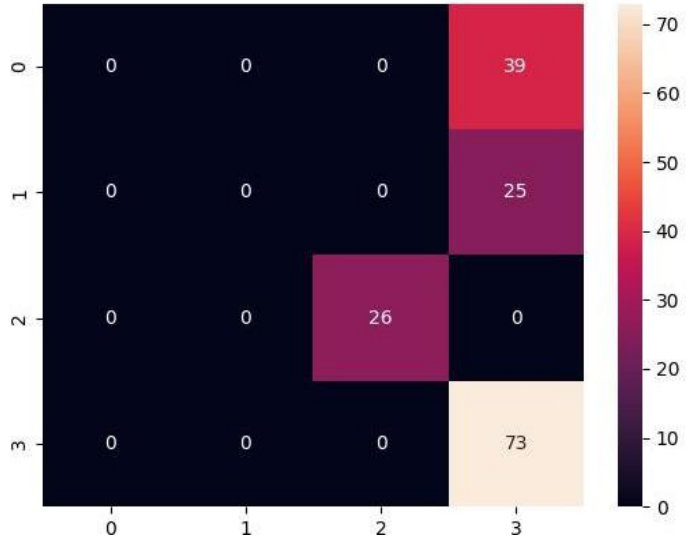

Fig 4: Confusion Matrix – Polynomial Kernel

The Polynomial model performs poorly, misclassifying most instances and heavily predicting the dominant class.

4.5 Confusion Matrix – RBF Kernel

The RBF SVM model's confusion matrix demonstrates excellent classification performance, with the majority of predictions accurately aligned along the diagonal. This shows that most cases in all classifications have been correctly classified.

The RBF model only generates a tiny number of off-diagonal values, which indicate slight misclassifications, in contrast to the Polynomial kernel. The findings show that some non-linear correlations in the data can be effectively captured by the RBF kernel. It does not, however, perform better than the Linear kernel, indicating that the dataset can be accurately classified without complicated adjustments.

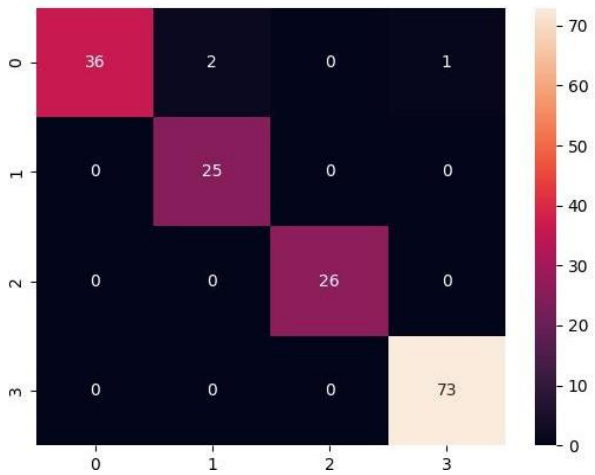

Fig 5: Confusion Matrix – RBF Kernel

5. Discussion:

The findings of this research underscore the impact of dataset properties on the performance of models. The elevated accuracy obtained from the Linear SVM suggests that the dataset is primarily linearly separable, indicating that job categories can be effectively differentiated using straightforward combinations of skill-based attributes. The dataset primarily consists of binary skill indicators, which seem to exhibit strong correlations with the target variable. This organized nature of the data simplifies the problem, enabling more straightforward models to achieve impressive results. On the other hand, the Polynomial kernel introduced excessive complexity, resulting in overfitting and inadequate generalization, especially due to the imbalance among the classes in the dataset.

The RBF kernel showed good results in identifying some non-linear patterns, but it fell short compared to the Linear model. In general, these results suggest that the choice of model should be based on the structure of the dataset, as simpler models may yield better performance than more complicated ones when the data is clearly defined.

6. Critical Evaluation:

Even though the Linear model reached perfect accuracy, this could suggest that the dataset is very organized and demonstrates strong connections between features and the target variable. Such circumstances facilitate class separation for the model, but achieving this level of performance is unlikely in more intricate, real-world situations where data is often noisy and unpredictable. The dataset heavily depends on binary skill-based features, which streamlines the classification process. While this is advantageous for modeling, it may not entirely reflect the complexity of actual job requirements, such as experience level, soft skills, or contextual elements.

In addition, the presence of **class imbalance** may influence model behaviour, particularly for more complex kernels, leading to biased predictions toward dominant classes. This is evident in the performance of the Polynomial model.

These factors highlight the importance of interpreting results carefully rather than relying solely on accuracy as a performance metric. A model that performs perfectly on structured data may not generalise well to unseen or more diverse datasets.

7. Conclusion:

This study looked at how different SVM models perform when classifying jobs using LinkedIn data. The results showed that the Linear model worked best and achieved perfect accuracy, meaning the data was easy to separate into categories using a straight line. The RBF model also worked well, but the Polynomial model performed worse because it was too complex and struggled with imbalanced data.

The results show that using a more complicated model does not always give better results. In this case, a simpler model worked better because the data was well-structured.

Overall, the study shows that choosing the right machine learning model depends on the type of data you have. It also reminds us to be careful when interpreting high accuracy, as good results on structured data may not work as well in real-world situations.

8.References

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